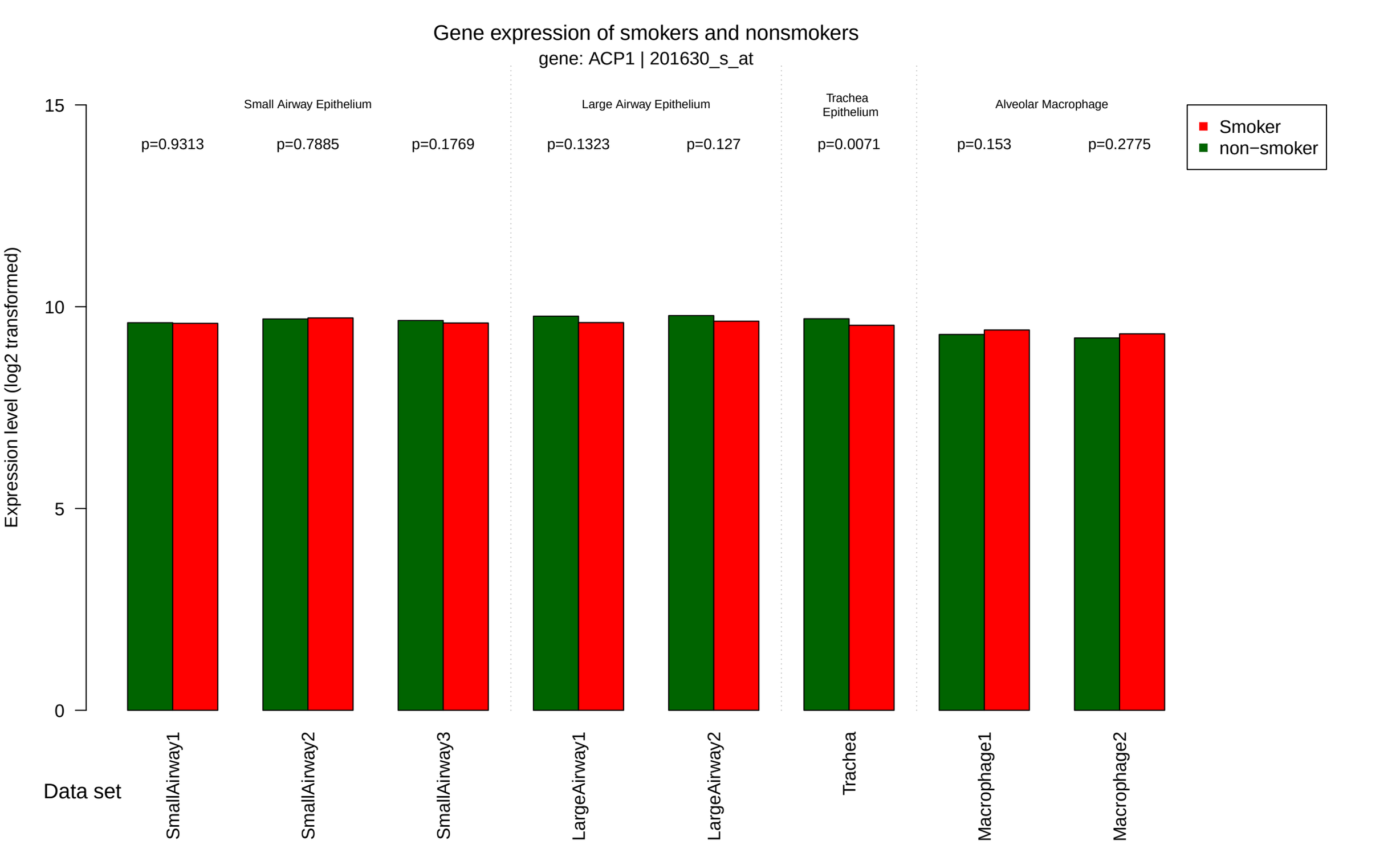
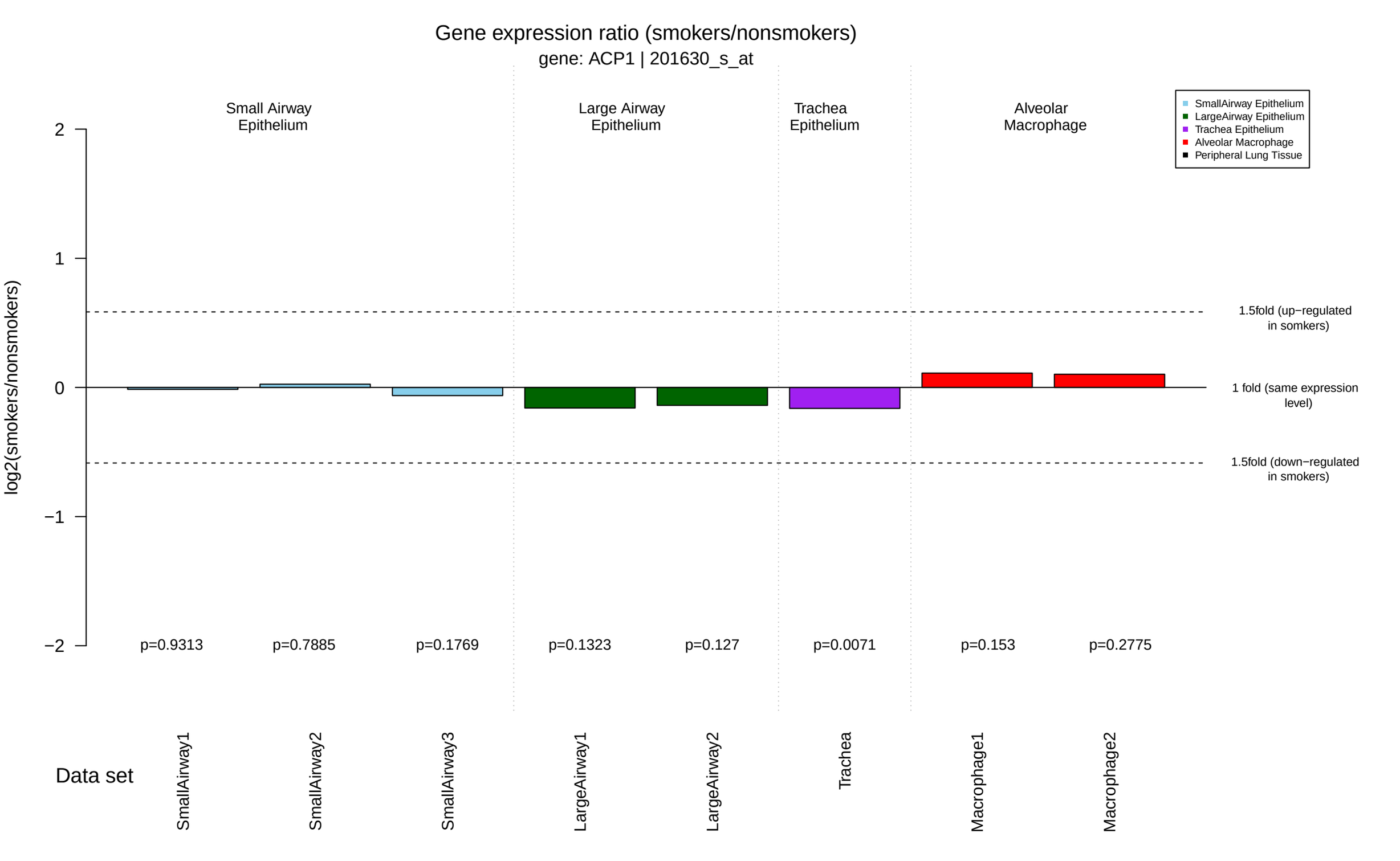
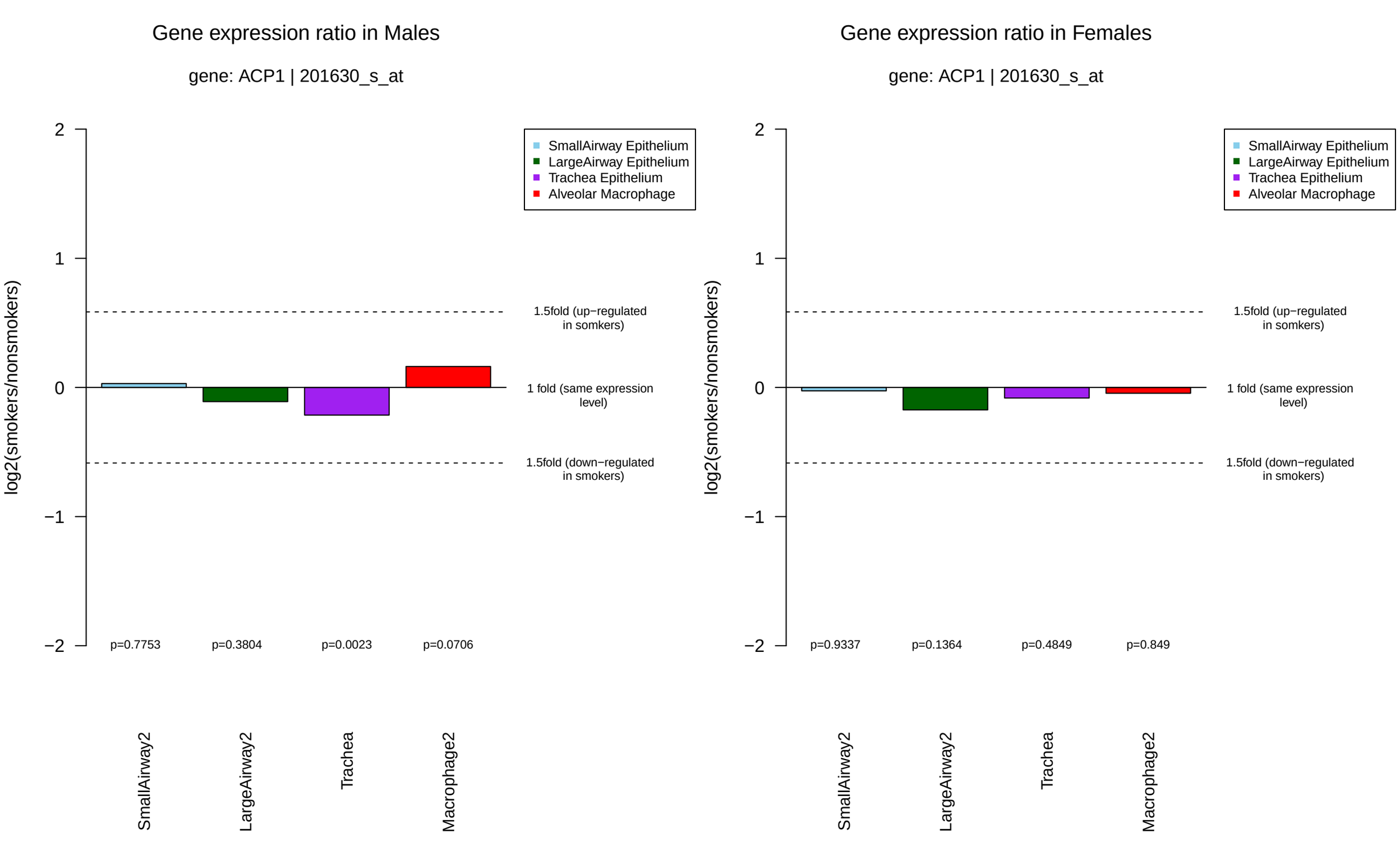
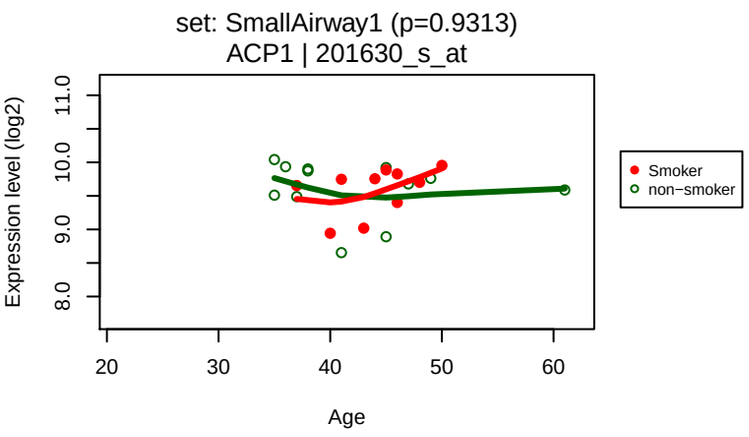
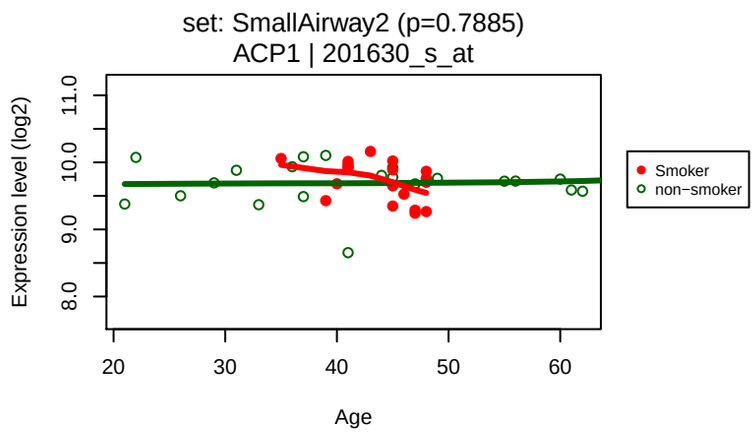
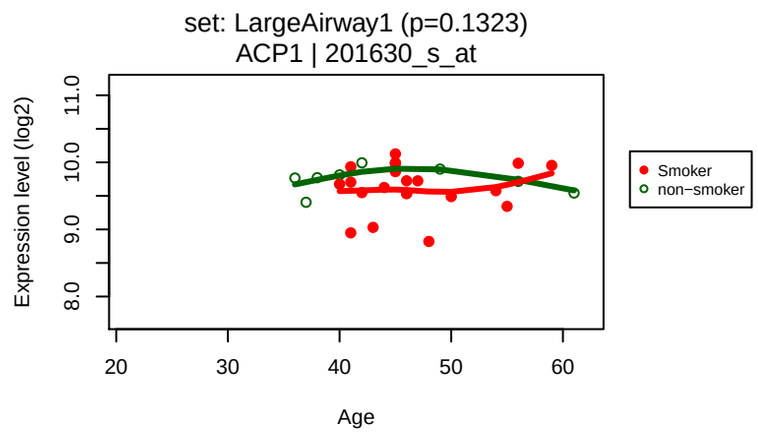

Gene expression of smokers and nonsmokers
gene: ACP1 | 201630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9313
p=0.7885
p=0.1769
p=0.1323
p=0.127
p=0.0071
p=0.153
p=0.2775
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

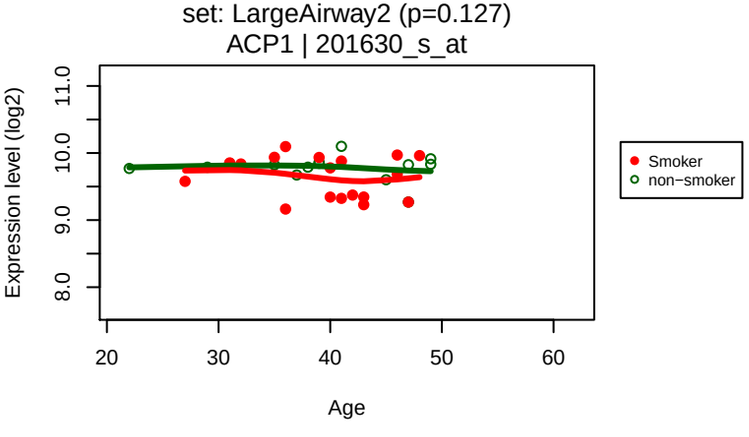

Gene expression ratio (smokers/nonsmokers)
gene: ACP1 | 201630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9313
p=0.7885
p=0.1769
p=0.1323
p=0.127
p=0.0071
p=0.153
p=0.2775
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

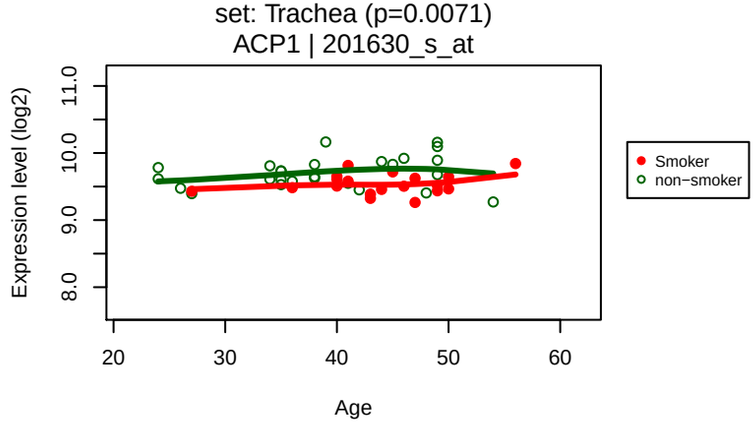

Gene expression ratio in Males
gene: ACP1 | 201630_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7753
p=0.3804
p=0.0023
p=0.0706
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACP1 | 201630_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9337
p=0.1364
p=0.4849
p=0.849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

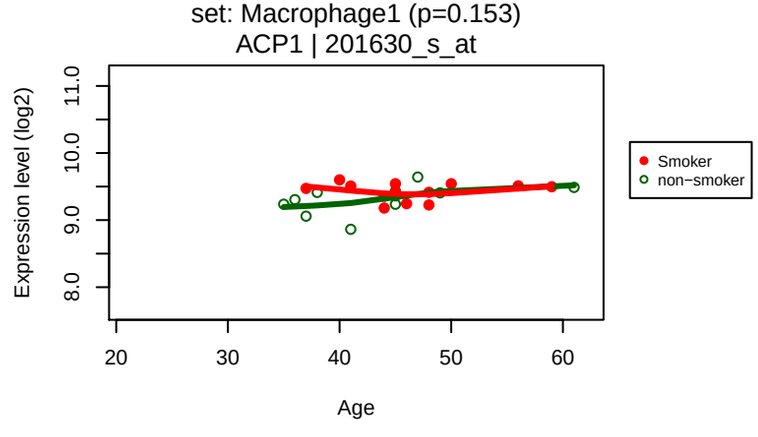

set: SmallAirway1 (p=0.9313)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

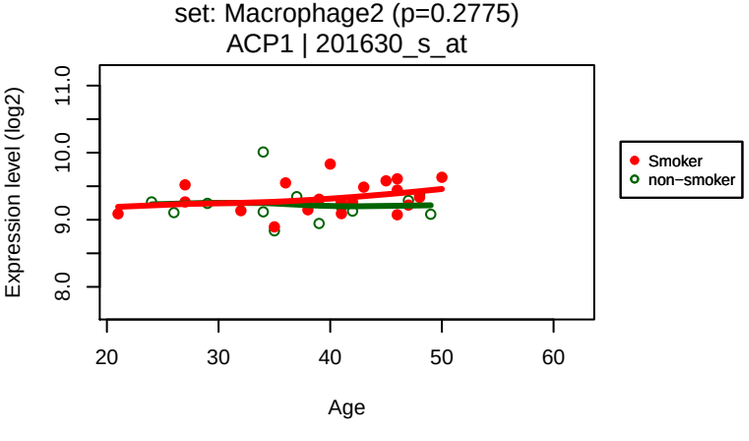

set: SmallAirway2 (p=0.7885)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

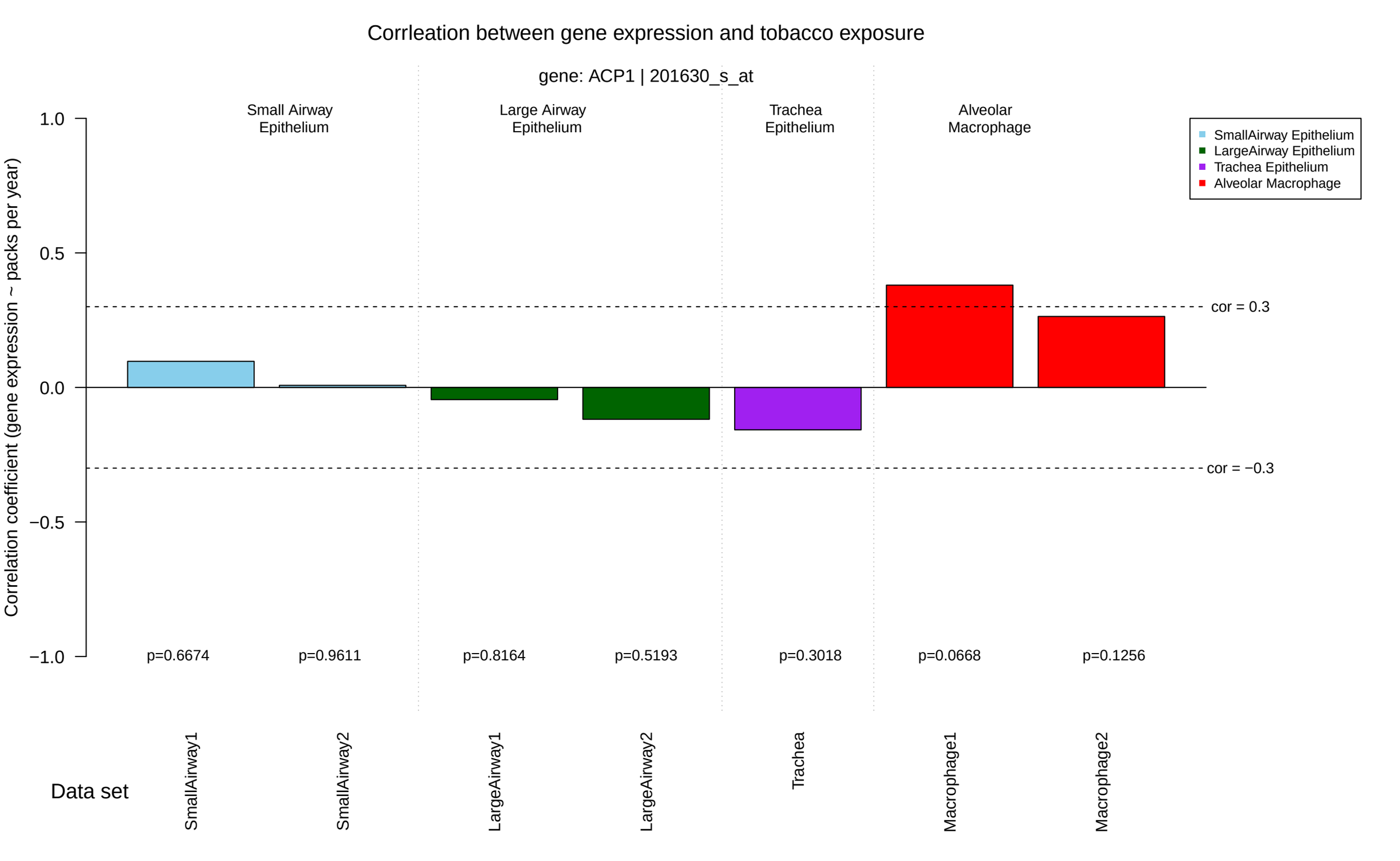

set: LargeAirway1 (p=0.1323)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.127)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0071)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.153)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2775)
ACP1 | 201630_s_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACP1 | 201630_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6674
p=0.9611
p=0.8164
p=0.5193
p=0.3018
p=0.0668
p=0.1256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2